

QUESTION 1

SOURCE CODE:

```
#include <iostream>
#include <string>
using namespace std;
struct car{
    string model;
    int year;
    float price;
};

int main(){
    car data[3];
    cout<<"Enter details of 3 cars"<<endl;
    for(int i=0;i<3;i++){
        cout<<"Car "<<i+1<<endl;
        cout<<"Model ";
        cin>>data[i].model;
        cout<<"Price ";
        cin>>data[i].price;
        cout<<"Year ";
        cin>>data[i].year;
    }
```

```

cout<<"Cars with price above 500000"<<endl;
float final_price;
for(int i=0;i<3;i++){
    final_price = data[i].price;
    if(final_price>500000){
        cout<<"Model " <<data[i].model<<endl;
        cout<<"Price " <<data[i].price<<endl;
        cout<<"Year " <<data[i].year<<endl;
    }
}
return 0;
}

```

OUTPUT:

```

Enter details of 3 cars
Car 1
Model Corolla
Price 340000
Year 2000
Car 2
Model Toyota
Price 505000
Year 2020
Car 3
Model Alto
Price 450000
Year 2019
Cars with price above 500000
Model Toyota
Price 505000
Year 2020

```

QUESTION 2:

SOURCE CODE:

```
struct address{
    string city;
    string country;
};
struct contact_info{
    string name;
    string phone_no;
    address add;
};
int main(){
    contact_info pple_data[10];
    for(int i=0;i<5;i++){
        cout<<"ENTER DETAILS OF PERSON "<<i+1<<endl;
        cout<<"Name: ";
        cin>>pple_data[i].name;
        cout<<"Phone Number ";
        cin>>pple_data[i].phone_no;
        cout<<"City ";
        cin>>pple_data[i].add.city;
        cout<<"Country ";
        cin>>pple_data[i].add.country;
    }
}
```

```
cout<<"====="<<endl;
for(int i=0;i<5;i++){
    cout<<"CONTACT INFORMATION OF PERSON "<<i+1<<endl;
    cout<<"Name: "<<ppl_data[i].name<<endl;

    cout<<"Phone Number "<<ppl_data[i].phone_no<<endl;

    cout<<"City "<<ppl_data[i].add.city<<endl;

    cout<<"Country "<<ppl_data[i].add.country<<endl;
}
}
```

OUTPUT:

ENTER DETAILS OF PERSON 1

Name: waleeja

Phone Number 0987654

City Peshawar

Country Pakistan

ENTER DETAILS OF PERSON 2

Name: Areej

Phone Number 05145675

City Peshawar

Country Pakistan

ENTER DETAILS OF PERSON 3

Name: Shawaiz

Phone Number 0345621

City Mumbai

Country India

ENTER DETAILS OF PERSON 4

Name: Mubashir

Phone Number 123654

City Toronto

Country Canada

ENTER DETAILS OF PERSON 5

Name: Wareesha

Phone Number 0981234

City Mansehra

Country Pakistan

=====

CONTACT INFORMATION OF PERSON 1

Name: waleeja

Phone Number 0987654

City Peshawar

Country Pakistan

CONTACT INFORMATION OF PERSON 2

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Phone Number 05145675

City Peshawar

Country Pakistan

CONTACT INFORMATION OF PERSON 3

Name: Shawaiz

Phone Number 0345621

City Mumbai

Country India

CONTACT INFORMATION OF PERSON 4

Name: Mubashir

Phone Number 123654

City Toronto

Country Canada

CONTACT INFORMATION OF PERSON 5

Name: Wareesha

Phone Number 0981234

City Mansehra

Country Pakistan

QUESTION 3:

SOURCE CODE:

```

struct camera{
    int megapixels;
};
struct mobile{
    camera cam;
    float price;
};
mobile info[5];

```

```

// //===== Logic=====

void show_report__mobile(){
int avg;
int sum=0;
int count1=0;
int count2=0;
int expensive;
int highest_MP;
// //===== All Data
for(int i=0;i<5;i++){
    cout<<"====="<<endl;
    cout<<"Data for mobile "<<i+1<<endl;
    cout<<"====="<<endl;
    cout<<"Megapixels: "<<info[i].cam.megapixels<<endl;
    cout<<"Price: "<<info[i].price<<endl;
}

// //===== Most expensive
expensive=info[0].price;
for(int i=0;i<5;i++){
    if(info[i].price>expensive){
        expensive=info[i].price;
    }
}
}

```

```

    }
    cout<<"Most Expensive Price: "<<expensive<<endl;

    highest_MP=info[0].cam.megapixels;
    for(int i=0;i<5;i++){
        if(info[i].cam.megapixels>highest_MP){
            highest_MP=info[i].cam.megapixels;
        }
    }
    cout<<"Highest Camera MP: "<<highest_MP<<endl;

    for(int i=0;i<5;i++){
        sum+=info[i].price;
        count1++;
    }
    avg=sum/count1;
    cout<<"Average Price: "<<avg<<endl;

```

```

int main(){

    for(int i=0;i<5;i++){
        cout<<"ENTER DATA FOR MOBILE "<<i+1<<" "<<endl;
        cout<<"Enter megapixels(>0): ";

        cin>>info[i].cam.megapixels;

        cout<<"Enter price(>=0): ";
        cin>>info[i].price;
        if(info[i].cam.megapixels<=0 || info[i].price<0){
            i--;
            cout<<"Invalid data"<<endl;
        }
    }
    show_report__mobile();
}

```

OUTPUT:


```
ENTER DATA FOR MOBILE 1
Enter megapixels(>0): 48
Enter price(>=0): 34000
ENTER DATA FOR MOBILE 2
Enter megapixels(>0): 52
Enter price(>=0): 50000
ENTER DATA FOR MOBILE 3
Enter megapixels(>0): 30
Enter price(>=0): 20000
ENTER DATA FOR MOBILE 4
Enter megapixels(>0): 46
Enter price(>=0): 30000
ENTER DATA FOR MOBILE 5
Enter megapixels(>0): 20
Enter price(>=0): 18500
```

```
=====
Data for mobile 1
=====
Megapixels: 48
Price: 34000
=====
Data for mobile 2
=====
Megapixels: 52
Price: 50000
=====
Data for mobile 3
=====
Megapixels: 30
Price: 20000
=====
Data for mobile 4
=====
Megapixels: 46
Price: 30000
=====
Data for mobile 5
=====
Megapixels: 20
Price: 18500
```

```
Most Expensive Price: 50000
Highest Camera MP: 52
Average Price: 30500
Mobiles with Camera >= 50MP: 1
```